



Geometry Art: Symmetry

A Complete Lesson Guide for Primary Students

Exploring symmetry in shapes, nature, and Indian art

Subject: Mathematics / Art

Duration: 60 minutes

Setting: Classroom

Learning Objectives

By the end of this lesson, students will be able to:

- **Define** symmetry in their own words.
- **Identify** symmetric and asymmetric objects in everyday life.
- **Draw** the line of symmetry on simple shapes.
- **Create** symmetric artwork using folding techniques.

Key Vocabulary

Symmetry

When both halves of a shape are exact mirror images of each other.

Half

One of two equal parts when a symmetric shape is folded.

Line of Symmetry

The fold line that divides a shape into two perfectly matching halves.

Lesson Structure (60 minutes)

Phase	Activity	Time	Slide(s)
Warm-Up	What Do You See? — Spot symmetric objects	5 min	Slide 2
Introduction	The Folding Secret — heart demo	5 min	Slide 3
Vocabulary	Key terms: Symmetry, Half, Line of Symmetry	5 min	Slide 4
Concepts	Symmetric vs Asymmetric examples with dotted fold lines	10 min	Slide 5
Practice	Spot the Symmetry! — call-and-response with class	5 min	Slide 6
Guided	Draw the line of symmetry on given shapes	5 min	Slide 7





Phase	Activity	Time	Slide(s)
Real-World	Find Symmetric Things at Home — discussion	5 min	Slide 8
Art Activity	Fold & Paint Butterfly OR Fold & Draw Mirror Kolam	15 min	Slides 9–11
Wrap-Up	Symmetry All Around Us — Taj Mahal, Rangoli, Peacock	3 min	Slide 12
Share	Gallery walk / show-and-tell	4 min	Slide 13
Exit Ticket	Name 1 symmetric + 1 non-symmetric object	3 min	Slide 14

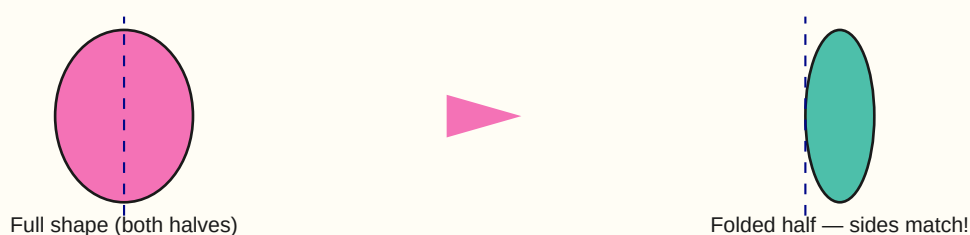


What Is Symmetry?

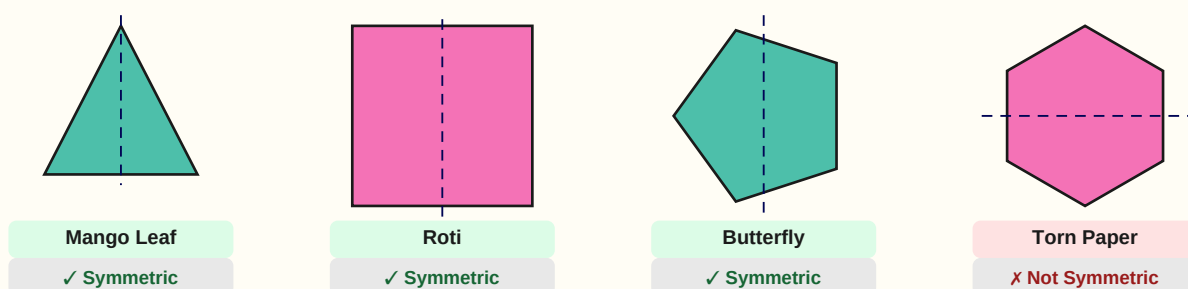
Symmetry occurs when a shape or object can be divided by a line so that both halves are **exact mirror images** of each other. This dividing line is called the **Line of Symmetry**.

Think of folding paper: if both halves match perfectly when you press them together, the shape has symmetry!

The Folding Secret — Demo



Symmetric and Asymmetric Shapes



The **dotted lines** show where a shape can be folded so both halves match perfectly. Torn paper has NO such line — it is **asymmetric**.

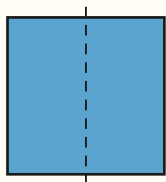
Everyday Symmetric and Asymmetric Objects

Object	Symmetric?	Lines	Why?
Butterfly	✓ Yes	1 (vertical)	Wings are mirror images of each other
Mango Leaf	✓ Yes	1 (vertical)	Both halves of the leaf match perfectly
Roti	✓ Yes	Many	Circular shape — infinite fold lines
Diya (lamp)	✓ Yes	1 (vertical)	Lamp body and flame are balanced
Human Face	✓ Yes	1 (vertical)	Eyes, ears, cheeks mirror each other
Torn Paper	✗ No	None	Irregular edges — no matching halves

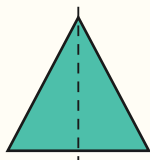


Object	Symmetric?	Lines	Why?
Squiggly Line	X No	None	Random curves — no fold line works

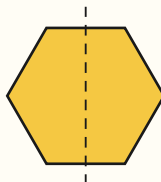
Lines of Symmetry in Common Shapes



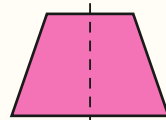
Square



Triangle



Hexagon



Trapezoid

Shape	Lines of Symmetry	Notes
Equilateral Triangle	3	Vertical + 2 diagonal lines
Square	4	2 through midpoints + 2 diagonal
Regular Hexagon	6	3 through vertices + 3 through midpoints
Circle	Infinite	Any diameter is a line of symmetry
Rectangle	2	Only through midpoints of opposite sides
Scalene Triangle	0	All sides different — no symmetry



Creative Art Activities

Option A — Fold & Paint: Rangoli Butterfly

1. Fold a sheet of paper in half vertically.
2. On ONE half, paint a design using bright colours.
3. While still wet, press both halves firmly together.
4. Slowly open to reveal the symmetric print!
5. Let dry, then add details or outlines.

Option B — Fold & Draw: Mirror Kolam

1. Draw a vertical centre line on your paper.
2. On the LEFT half, draw part of a kolam pattern.
3. Mirror each dot and curve on the RIGHT half.
4. Unfold / check — both sides should match perfectly.
5. Colour your kolam with bright traditional colours.

Materials — Option A: White paper, poster or finger paint, paintbrush, newspaper to protect desks.

Materials — Option B: White or grid paper, pencil, ruler, coloured pencils or crayons.

Symmetry in Indian Culture and Nature

Example	Where to See It	Lines of Symmetry
Taj Mahal	Agra, Uttar Pradesh	1 — reflected in the pool
Peacock Feather	Indian national bird	1 — central spine
Rangoli	Festival floors & doorsteps	4 or more — complex patterns
Lotus Flower	National flower of India	Multiple radial lines
Roti	Kitchen — everyday bread	Approximate circular symmetry
Human Face	Look in the mirror!	1 — vertical

Teacher Tips and Differentiation

For Learners Who Need Support	For On-Level Learners	For Advanced Learners
• Use physical paper folding before abstract shapes. • Provide pre-drawn shapes for colouring halves. • Focus on 1 line of symmetry only.	• Try identifying lines of symmetry on 3-D objects. • Complete both art activity options (A and B). • Find 5 symmetric things at home (homework).	• Explore rotational symmetry in patterns. • Count all lines of symmetry in a regular polygon. • Research the mathematics of traditional Kolam art.

Exit Ticket — What Did You Learn?



1. Name ONE symmetric object you know.	Answer: _____
2. Name ONE object that is NOT symmetric.	Answer: _____
3. Where do you draw the line of symmetry on a triangle?	Answer: _____
4. True or False: A squiggly line has symmetry.	Answer: _____

Answer Key and Assessment Rubric

Exit Ticket Answers:

1. Symmetric examples: butterfly, roti, mango leaf, diya, face, lotus, Taj Mahal.
2. Non-symmetric examples: torn paper, squiggly line, irregular cloud.
3. A vertical line from the top vertex to the midpoint of the base.
4. False — a squiggly line has no line of symmetry.

Criterion	3 — Exceeds	2 — Meets	1 — Approaching
Identifies symmetry	Correctly identifies all symmetric & asymmetric objects	Identifies most with minor errors	Needs support to tell them apart
Line of symmetry	Draws correct line(s) on multiple shapes	Draws correct line on at least one shape	Attempts but places line incorrectly
Vocabulary	Uses all three key terms accurately	Uses at least two terms correctly	Uses one term or needs prompting
Art activity	Creates clearly symmetric artwork independently	Creates artwork with noticeable symmetry	Participates with teacher support

Geometry Art: Symmetry — Lesson Guide | Aligned to primary mathematics curriculum | Every child is an artist and a mathematician!

